Solve with us 3.1 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

Definition of a vector space V: Definition of a vector space V:

- A vector space V over R is a set along with two functions:

1. $+: V \times V \to V$ (V is closed under vector addition i.e., if $x, y \in V \implies x + y \in V, \ \forall \ x, y \in V$)
2. $.: R \times V \to V$ (V is closed under scalar multiplication i.e., if $x \in V$ and $c \in R \implies c.x \in V, \ \forall \ x \in V$, and $\forall c \in R$)

- Axioms of vector additions of a vector space

1. $v_1 + v_2 = v_2 + v_1, \ \forall \ v_1, v_2 \in V$.
2. $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \ \forall \ v_1, v_2$ and $v_3 \in V$.
3. There exists an element 0 (called the zero vector of V) in V such that $0 + v = v, \ \forall \ v \in V$.
4. There exists a vector v' in V such that $v' + v = v + v' = 0, \ \forall \ v \in V$ (v' is called the inverse (or negative) of v with respect to addition).

- Axioms relating vector addition and scalar multiplication of a vector space (since this is usual multiplication in real numbers, we suppress the "." sign).

1. For each vector $v \in V$, $1v = v$.
2. For each vector $v \in V$ and for each pair $a, b \in R, (a + b)v = av + bv$.
3. For each $a \in R$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$.
4. For each vector $v \in V$ and for each pair $a, b \in R, (ab)v = a(bv)$.

Consider a set $V = \{(1, -y) \mid y \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2. Answer the following questions (1) and (2).

1 point
Which of the following option(s) is (are) true?

[Hint:Hint: Check the axioms of vector space.]

- [ ] $(1, 2) \in V$.
- [ ] $(2, 3) \in V$.
- [ ] $V = R^2$.
- [ ] $v_1 = (1, -5)$ and $v_2 = (1, 7) \in V \implies v_1 + v_2 \in V$.
- [ ] $v = (1, 9) \in V, c \in R \implies cv \in V, \ \forall \ c \in R$.

☐

For all $v \in V, cv \in V$ For all v ∈ V , cv ∈ V only for $c = 1$c = 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(1, 2) \in V$(1, 2) ∈ V .

For all $v \in V, cv \in V$ For all v ∈ V , cv ∈ V only for $c = 1$c = 1.

1 point

Which of the following option(s) is (are) true?

☐

Zero element does not exist in VV .

☐

Zero element of VV is (0,0).

☐ Inverse element of (1, 5) is (-1, -5).

☐ Inverse element of (1, 5) is (1, -5).

☐ Inverse element of (1, 5) does not exist.

☐

Inverse element of any element of VV does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Zero element does not exist in VV .
Inverse element of (1, 5) does not exist.
Inverse element of any element of VV does not exist.

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ V = {(x, y, z) | x + y + z = 0, x, y, z ∈ R} ⊆ R3 with the usual addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$, (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2),

where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ where (x1, y1, z1), (x2, y2, z2) ∈ V and

scalar multiplication: $c(x, y, z) = (cx, cy, cy), \quad c \in R$ and $(x, y, z) \in V$.c(x, y, z) = (cx, cy, cy), c ∈ R and (x, y, z) ∈ V .

Answer the following questions (3) and (4).

1 point

Which of the following option(s) is (are) true?

[Hint:Hint: For a vector $v = (x, y, z)$v = (x, y, z), check whether the condition $x + y + z = 0$ x + y + z = 0 is satisfied or not.]

☐

(1, 2, -3) $\in V$∈ V .

☐

(-1, 2, 1) $\in V$∈ V .

☐

$V = R^3$ V = R3.

☐

$(5, -7, 2) \in V$(5, −7, 2) ∈ V .

No, the answer is incorrect.
Score: 0
Accepted Answers:

(1, 2, -3) $\in V$∈ V .

$(5, -7, 2) \in V$.

1 point

Inverse element of (-2, -4, 6) with respect to vector addition is

○ (2, 4, -6).
○ (2, -4, -6).
○ (2, 4, 6).
○ (-2, 4, -6).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(2, 4, -6).

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ with the usual addition as in R^3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ \left(\frac{cx}{2}, \frac{y}{c}, cz\right) & c \neq 0 \end{cases} \quad (x, y, z) \in V,\ c \in R$$

Which of the following option(s) is (are) true?

[Hint: By adding any two elements in V, or by multiplying a scalar with any element in V, check whether the resultant element is in V or not.]

☐ V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
☐ V is closed under scalar multiplication i.e., $(c \in R, v \in V \implies cv \in V)$.
☐ V has zero element with respect to addition.
☐ $1.v = v$, where $1 \in R$ and $v \in V$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
V has zero element with respect to addition.

1 point

Consider usual addition and scalar multiplication on R^2 and the subset $V = \{(1, -y) \mid y \in R\} \subseteq R^2$.

Which of the following option(s) is (are) true?

[Hint: Take two arbitrary elements $(1, -y_1)$ and $(1, -y_2)$ from V and check the conditions given in the options.]

☐ For each element $v \in V$ and for each pair $a, b \in R$, $(a + b)v = av + bv$.
☐ For each $a \in R$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$

) = av1 + av2.

☐ For each element $v \in V$ and for each pair $a, b \in R, (ab)v = a(bv)$.

☐ The set V with the usual addition and scalar multiplication in R^2 is a vector space.

No, the answer is incorrect.
Score: 0
Accepted Answers:

For each element $v \in V$ and for each pair $a, b \in R, (a+b)v = av + bv$.

For each $a \in R$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$.

For each element $v \in V$ and for each pair $a, b \in R, (ab)v = a(bv)$.

1 point

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in R\} \subseteq R^3$ with the usual addition as in R^3 and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0,0,0) & c = 0 \\ \left(\frac{cx}{2}, \frac{y}{c}, cz\right) & c \neq 0 \end{cases} \quad (x, y, z) \in V,\ c \in R$$

Which of the following option(s) is (are) true?

○ For each element $v \in V$ and for each pair $a, b \in R, (a+b)v = av + bv$

○ For each $a \in R$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$

○ For each element $v \in V$ and for each pair $a, b \in R, (ab)v = a(bv)$

○ V is a vector space.

No, the answer is incorrect.
Score: 0
Accepted Answers:

For each $a \in R$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$

Key Points: 2

- Definition of a subspace: A subset W of a vector space V is a subspace of V if it is a vector space with respect to the same operations defined on V.
- Criteria to check a subset W of a vector space V is a subspace:

1. Zero vector should be in W i.e., $0 \in W$.
2. If two vectors v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
3. Let $c \in R$ and if $v \in W$, then $cv \in W$

1 point

Let $W = \{(x, y) \mid x + 3y = 1, x, y \in R\} \subseteq R^2$. Which of the following options are true?

☐

Zero element exists in W.

☐

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.

☐

Let $c \in R$ and if $v \in W$, then $cv \in W$

☐

W is not a subspace of R^2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.

W is not a subspace of R^2.

1 point

Let $W = \{A \mid A \in M_{2\times2}(R) \text{ and } A \text{ is a diagonal matrix}\}$ be a subset of a vector space $M_{2\times2}(R)$. Which of the following option(s) is (are) true?

☐

Zero vector does not exist in W.

☐

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.

☐

Let $c \in R$ and if $v \in W$, then $cv \in W$

☐

W is a subspace of $M_{2\times2}(R)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.

Let $c \in R$ and if $v \in W$, then $cv \in W$

W is a subspace of $M_{2\times2}(R)$.

Key Points: 3

- Linear combination: Let V be a vector space and $v_1, v_2, ...v_k \in V$. The linear combination of $v_1, v_2, ...v_k$ with coefficients $a_1, a_2, ...a_k \in R$ is the vector $v = \sum_{i=1}^{k} a_i v_i$.
- A vector $v \in V$ is said to be a linear combination of $v_1, v_2, ...v_k \in V$ if there exist $a_1, a_2, ...a_k \in R$ so that $v = \sum_{i=1}^{k} a_i v_i$

1 point

Which of the following vectors are linear combinations of (2, 1) and (6, 3) with usual addition and scalar multiplication in R^2?

☐ (2, 1) + 5(6, 3)

☐ (2, 1)

☐ (6, 3)

☐ (3, 6)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(2, 1) + 5(6, 3)
(2, 1)
(6, 3)

1 point

Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ *i.e.*, $v_1 = av_2 + bv_3 + cv_4$, then which of the following options are true?

☐ If $a \neq 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.

☐ If $a = 0$ and $b \neq 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.

☐ v_2 is a linear combination of v_2.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $a \neq 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.
If $a = 0$ and $b \neq 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.
v_2 is a linear combination of v_2.